



Mobility System Design and Testing for GREMLINS Microrovers



Avery Agnes
Jessica Gogol

Adish Katwal
Shawn Mariner

Aryan
William Murgolo

Dr. David Akin
Professor/Lab Director

Background

The GREMLINS project develops multi-rover teams to autonomously build lunar infrastructure, including clearing landing areas, digging foundations, and constructing berms using regolith. Rovers must be small and lightweight, so teamwork is essential for large-scale tasks. Our team uses small-scale microrovers on Earth to test how different mobility designs like wheels vs. treads and various grouser shapes will affect performance on slopes and rough terrain.

Summary of Research

The research compared small rover treads and wheels to determine which offers better traction and mobility on loose lunar regolith, testing different grouser shapes and measuring slippage, distance, and speed during climbs. Wheels were expected to outperform treads, with chevron grousers providing the best traction. The findings will help inform the design of scalable GREMLINS microrovers for autonomous lunar construction.

Methodology

- Designed and 3D printed wheel and tread prototypes in CAD
- Tested 6 grouser shapes: Chevron, straight, spikes, angled lines, semicircle, and double chevron
- Conducted sandbox tests at slopes of 0°, 5°, 10°, and 15° with rover loads of (0 lb, 1 lb, 2 lb, 3 lb)
- Measure the traction, slippage, climb success, and average speed for each configuration
- Recorded data using graphs, photos, and video to compare designs
- Iteratively adjusted designs throughout wheel development process



Figure 1: Testing rover mobility on slopes with varying weights

Hypothesis / Objectives

- Wheels expected to outperform treads in overall mobility and speed
- Hypothesis: Chevron grousers predicted to provide best traction and stability on loose regolith
- Goal: Identify the optimal combination of mobility system and grouser design for lunar microrovers

Conclusion / Outcome

- Testing currently in progress with multiple slope angles and rover weights
- Preliminary data shows wheels perform slightly better on moderate slopes, while treads excel on very loose sand
- Final analysis will determine if Chevron grousers outperform other designs
- Results will guide GREMLINS rover design for scalable lunar construction

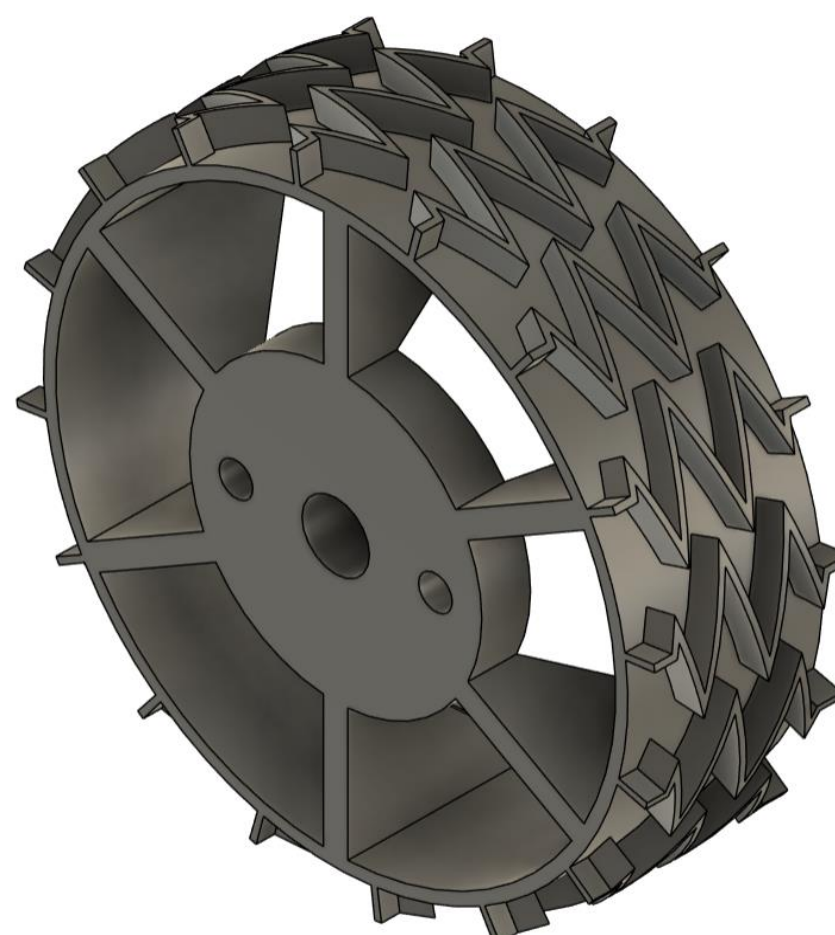


Figure 2: CAD model of a Double Chevron grouser design

Teamwork & Takeaways

- Team worked on tasks like design, prototyping, testing, and data analysis
- Collaborated weekly in the Neutral Buoyancy Research Facility to plan and conduct experiments
- Practiced iterative design: build → test → analyze → redesign
- Learned how mobility system design affects rover performance on slopes and loose terrain
- Findings will inform future GREMLINS rover designs for autonomous lunar construction

Random photo dump